PRACTICAL-01

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Section-B4

Batch-B2

SEPM LAB-1

What is UML 2?

UML 2 (Unified Modeling Language Version 2) is a **standard visual modeling language** used to **analyze, design, visualize, and document** software systems before coding begins. It is maintained by the **Object Management Group (OMG)** and is independent of any programming language. UML 2 expanded earlier versions with richer structural and behavioral modeling, adding a total of **13 standard diagram types**.

Why UML 2 is used?

- Understand system requirements
- Design software architecture
- Improve communication among developers, testers, and clients
- Reduce development errors
- Document software for future maintenance

Different Tool

1. StarUML

Description

A popular UML modeling tool used to create UML 2 diagrams.

Features

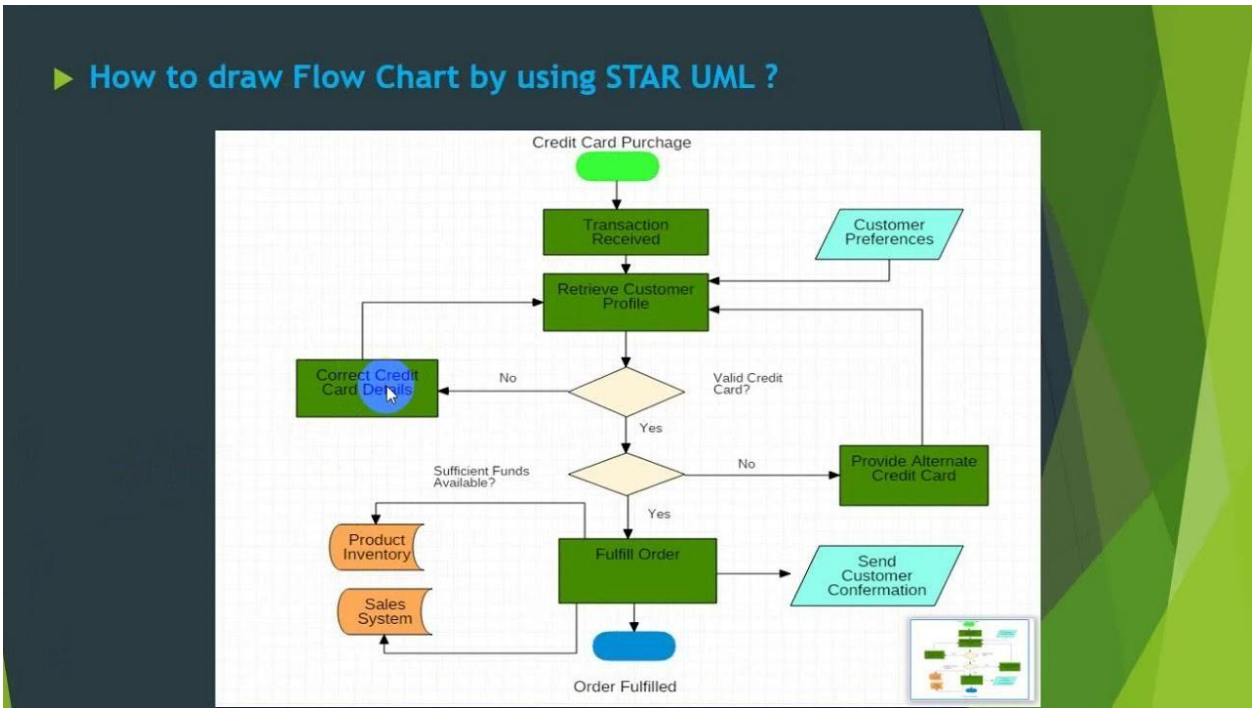
- Supports all UML diagrams
- Code generation
- Reverse engineering
- Export to PDF and images

Advantages

- Easy interface
- Fast diagram creation
- Lightweight

Used for

- Class Diagram
- Use Case Diagram
- Sequence Diagram
- Activity Diagram



2. Visual Paradigm

Description

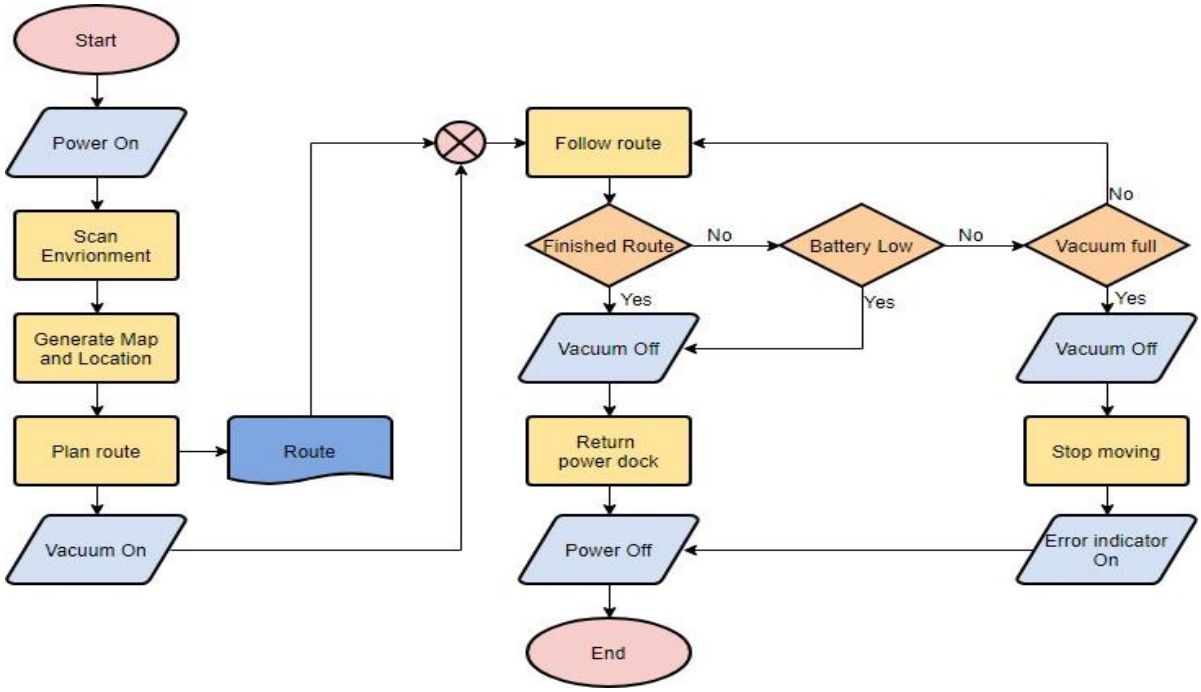
A professional software design tool supporting UML, BPMN, ERD, and Agile development.

Features

- Team collaboration
- Database design
- Requirement management
- Agile project support

Advantages

- Complete software development support
- Excellent documentation
- Project planning features



3. Lucidchart

Description

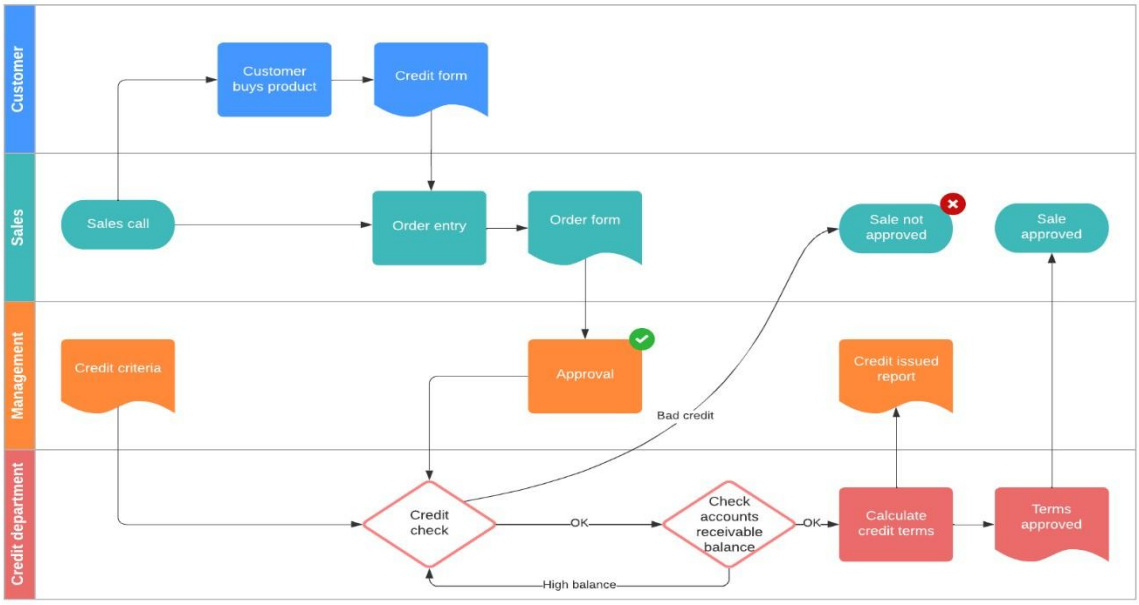
A web-based diagramming application.

Features

- Online collaboration
- Cloud storage
- Drag-and-drop interface

Advantages

- No installation required
- Multiple users can edit simultaneously
- Best for:
- Distributed software teams
- Project documentation



4. Draw.io (diagrams.net)

Description

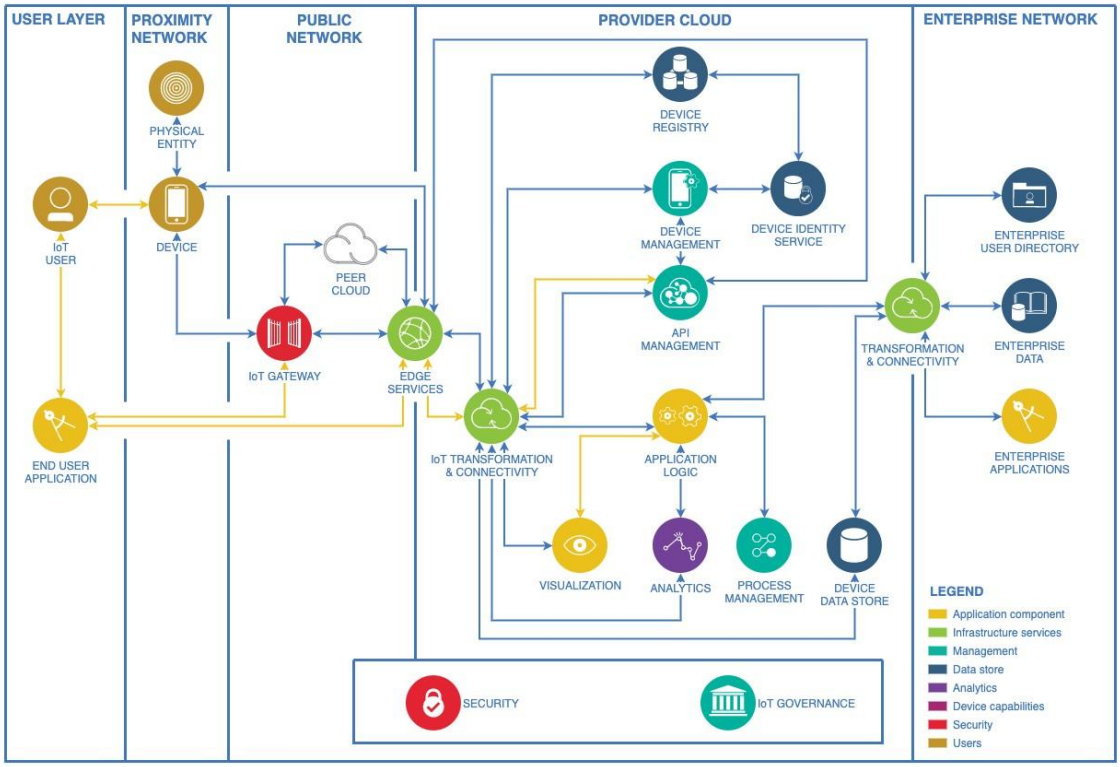
a. A free online diagram creation tool.

Features

- UML diagrams
- Flowcharts
- ER diagrams
- Network diagrams

Advantages

- Completely free
- Works offline and online • Saves to Google Drive or OneDrive
- Suitable for:
- College projects
- Assignments



5. Microsoft Visio

Description

Microsoft's professional diagramming software.

Features

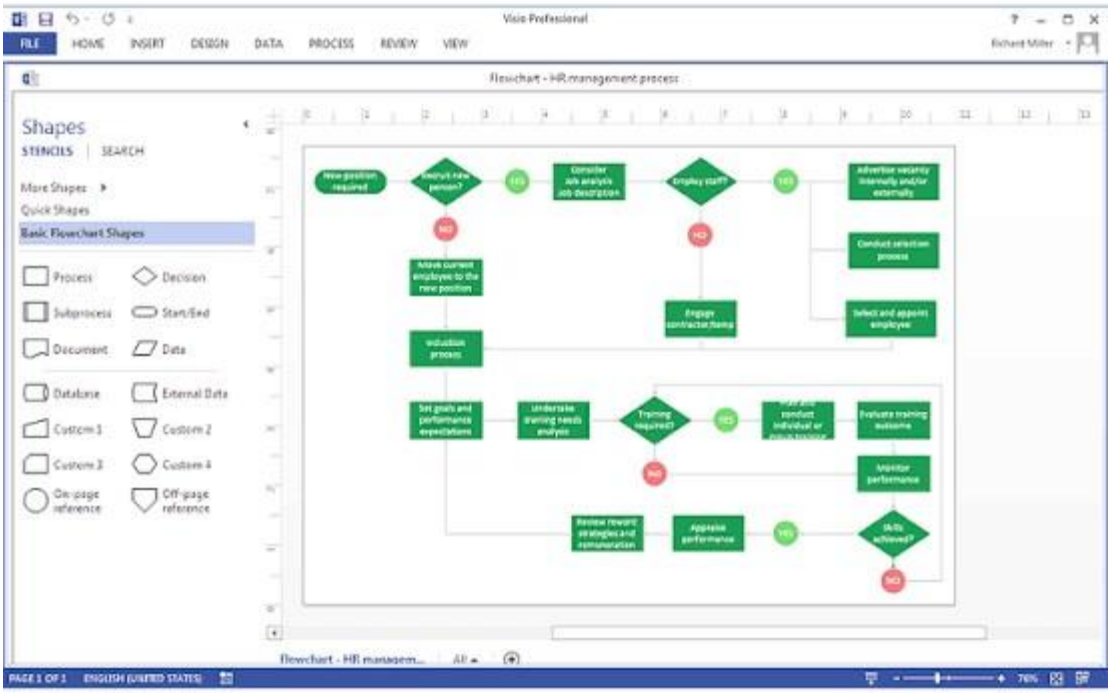
- UML templates
- Network diagrams
- Business process modeling

Advantages

- Professional quality diagrams
- Integrates with Microsoft Office

Used in:

- Enterprises
- IT companies



6. Enterprise Architect

Description

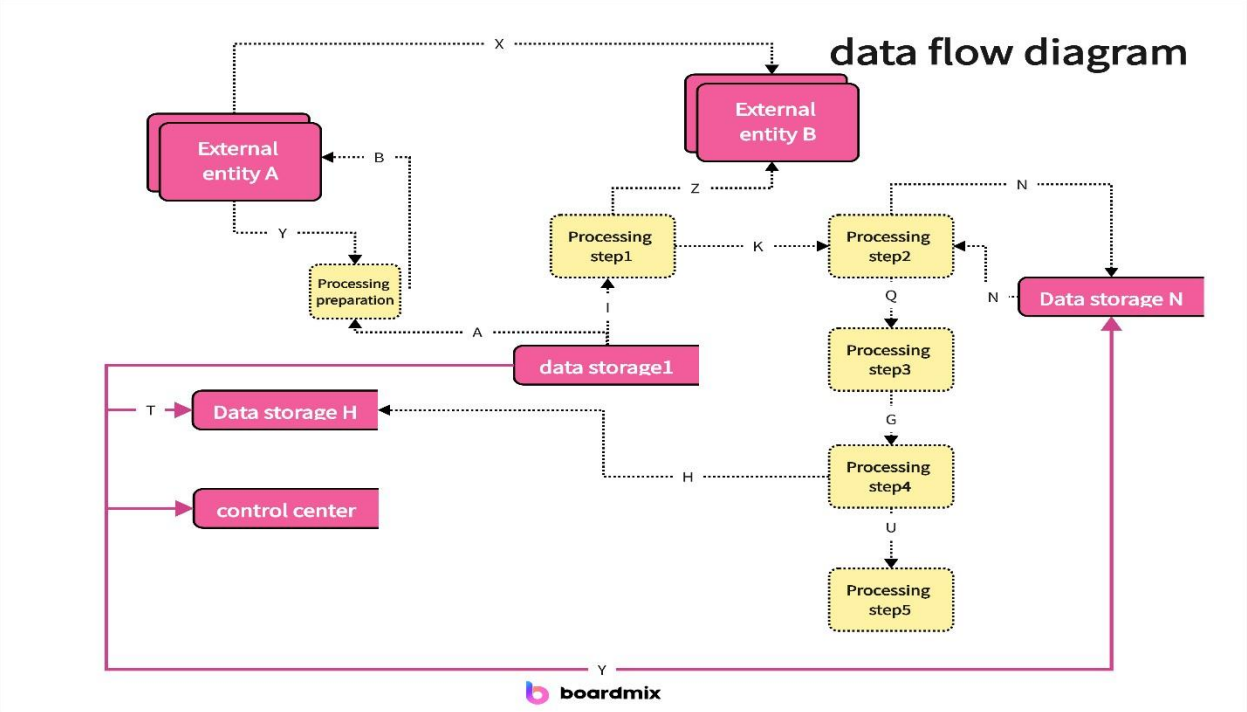
An advanced modeling platform for enterprise software projects.

Features

- UML 2 support
- Database modeling
- Requirement management
- Code engineering
- Team collaboration

Advantages

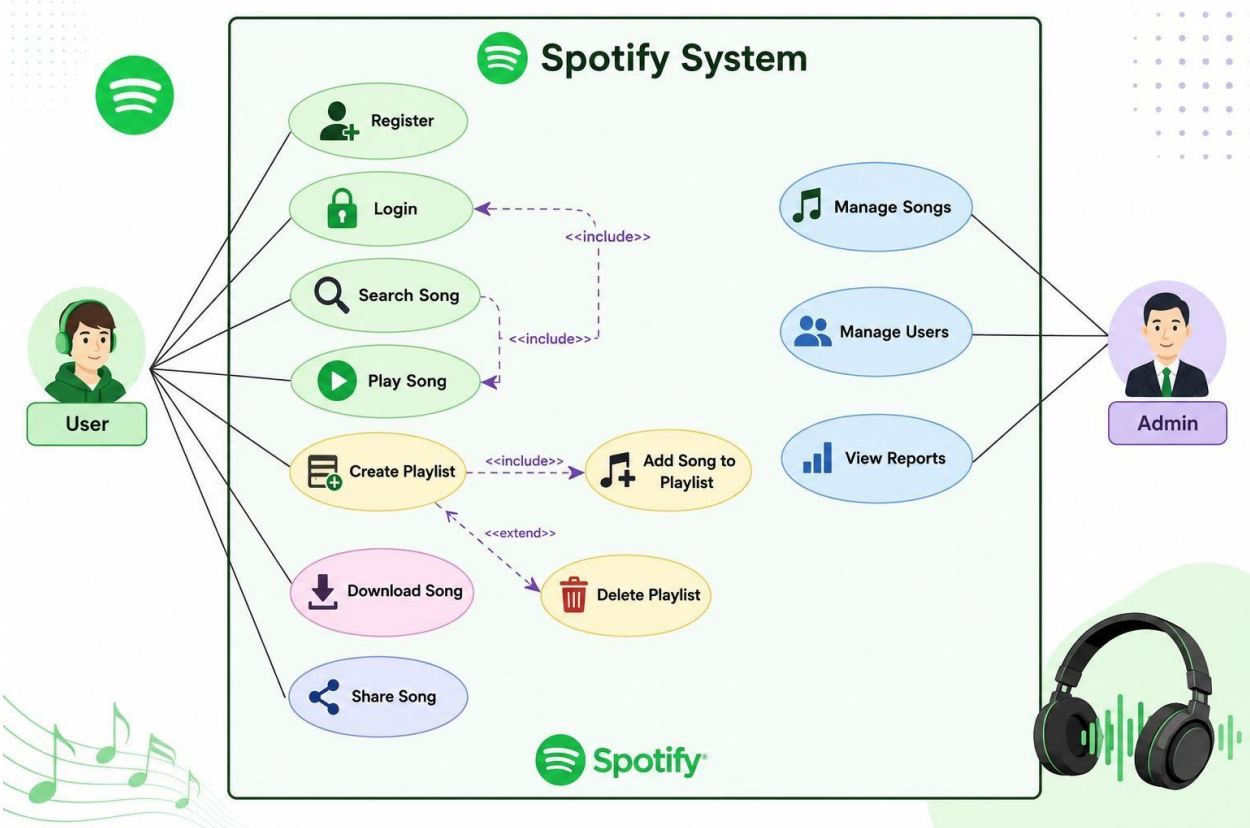
- Handles large and complex projects
- Strong documentation support
- Used by:
- Large software companies
- Government projects



SPOITFY USE CASE DIAGRAM

Observation:

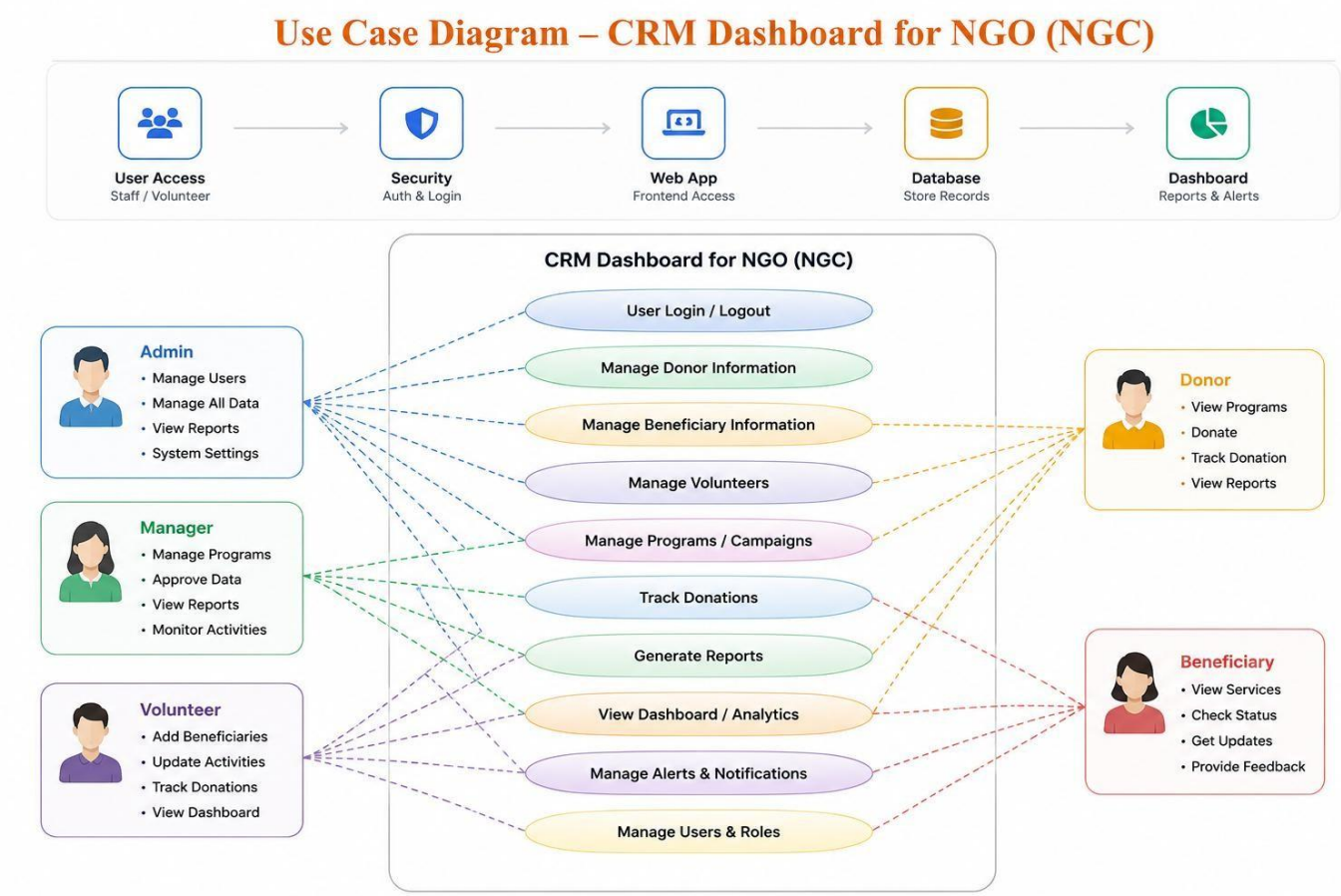
The given use case diagram represents a music application and shows the interaction between two actors: User and Admin. The user can perform activities such as downloading songs, checking song quality, creating playlists, playing songs, becoming a premium member, and viewing song information. The admin is responsible for managing users, artists, and advertisements. The diagram also uses «include» relationships to indicate mandatory features, such as checking song quality while downloading songs and providing additional features like Listen Together and View Connections for premium members. The «extend» relationship shows that Play Next is an optional feature available while playing a song. Overall, the diagram clearly defines the system functionalities and the responsibilities of both the user and the admin, making the system requirements easy to understand.



CEP USE CASE DIAGRAM

Observation:

The given use case diagram represents the CRM Dashboard for an NGO, showing how different actors interact with the system. The main actors are Admin, Volunteer, Donor, Beneficiary, and Manager, each having specific responsibilities based on their roles. The admin manages users, donors, volunteers, beneficiaries, and generates reports. The Volunteer updates activities and manages beneficiary information, while the Donor can register, make donations, and view donation history. The Beneficiary can view services and submit requests, and the Manager monitors overall activities and analyzes reports. The diagram also includes «include» and «extend» relationships to represent mandatory and optional functionalities. Overall, the use case diagram clearly defines the roles, responsibilities, and interactions within the CRM Dashboard, making the system easy to understand and ensuring efficient management of NGO operations.



➢ Software Engineering and Project Management Tools

1. Jira

Definition

Jira is a project management and issue-tracking tool developed by Atlassian. It is mainly used for Agile software development to plan, assign, and track project tasks.

Features

- Sprint Planning
- Scrum and Kanban Boards
- Bug Tracking
- Task Assignment
- Progress Reports
- Project Dashboard

Advantages

- Improves team collaboration.
- Tracks project progress in real time.
- Supports Agile and Scrum methodologies.

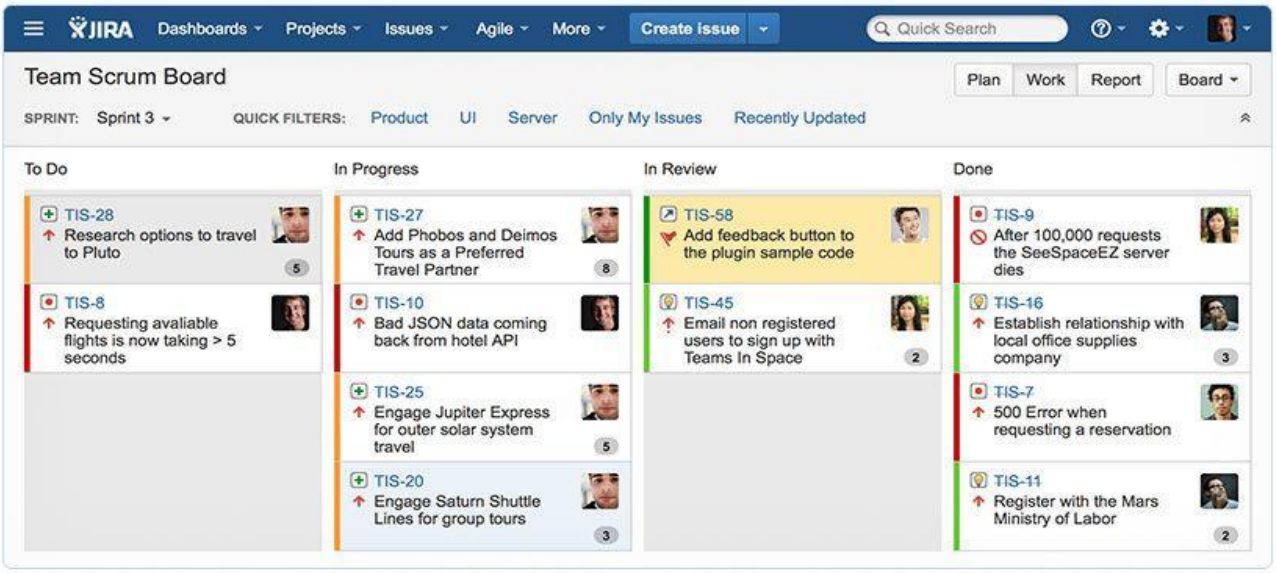
- Easy task and bug management.

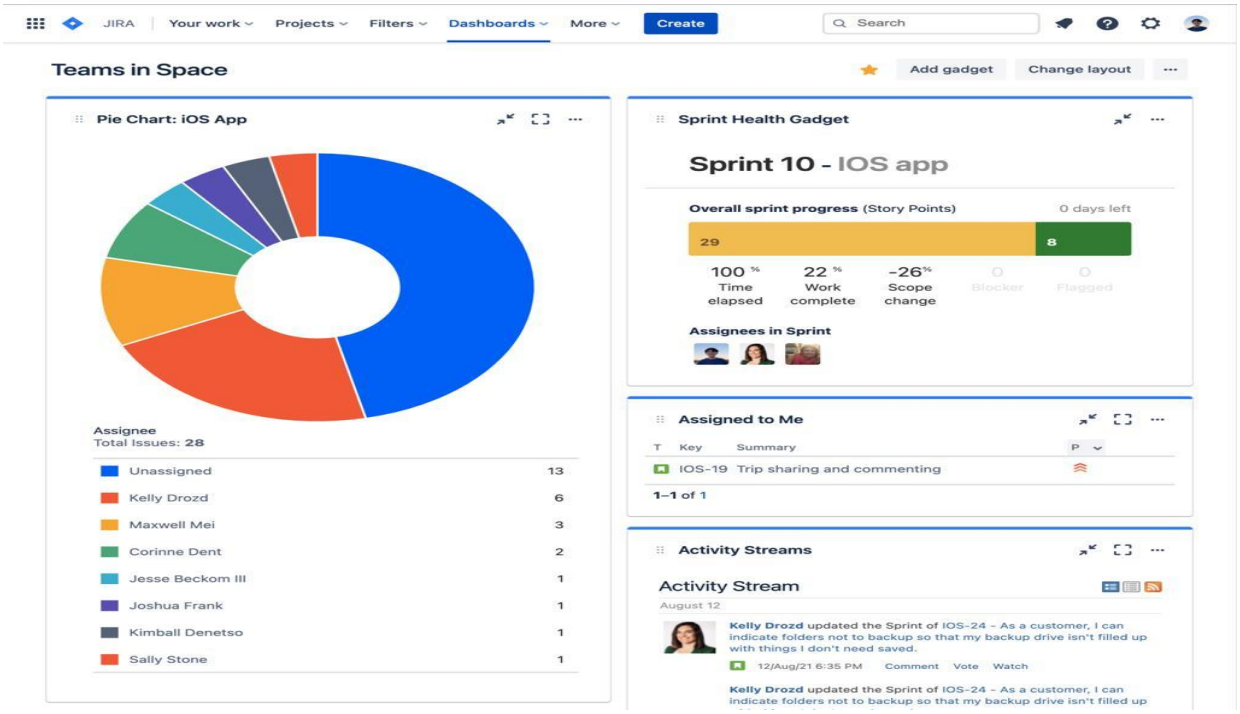
Disadvantages

- Can be difficult for beginners.
- Advanced features require a paid plan.

Real-World Example

A software company developing a food delivery app uses Jira to create user stories such as *User Login*, *Online Payment*, and *Order Tracking*. Team members update the status from To Do → In Progress → Done, allowing the project manager to monitor progress.





2. Microsoft Project (MS Project)

Definition

Microsoft Project is a project management tool used to **plan, schedule, and monitor projects**. It helps project managers estimate project duration, allocate resources, and track progress.

Features

- Gantt Charts
- Task Scheduling
- Resource Management
- Budget Planning
- Milestone Tracking
- Progress Monitoring

Advantages

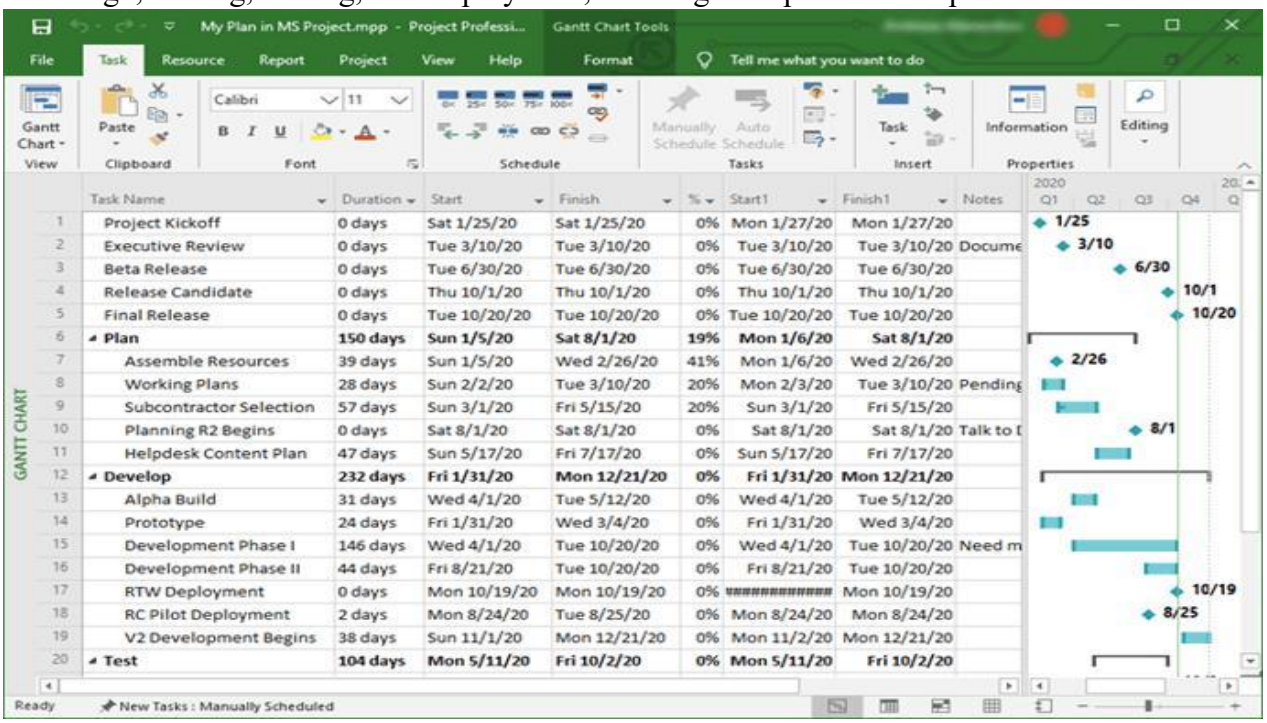
- Excellent project scheduling.
- Effective resource allocation.
- Generates professional reports.
- Helps manage project deadlines.

Disadvantages

- Paid software.
- Requires training to use effectively.

Real-World Example

A company developing an **E-commerce website** uses MS Project to schedule tasks like Requirement Analysis, UI Design, Coding, Testing, and Deployment, ensuring each phase is completed on time.



3. GitHub

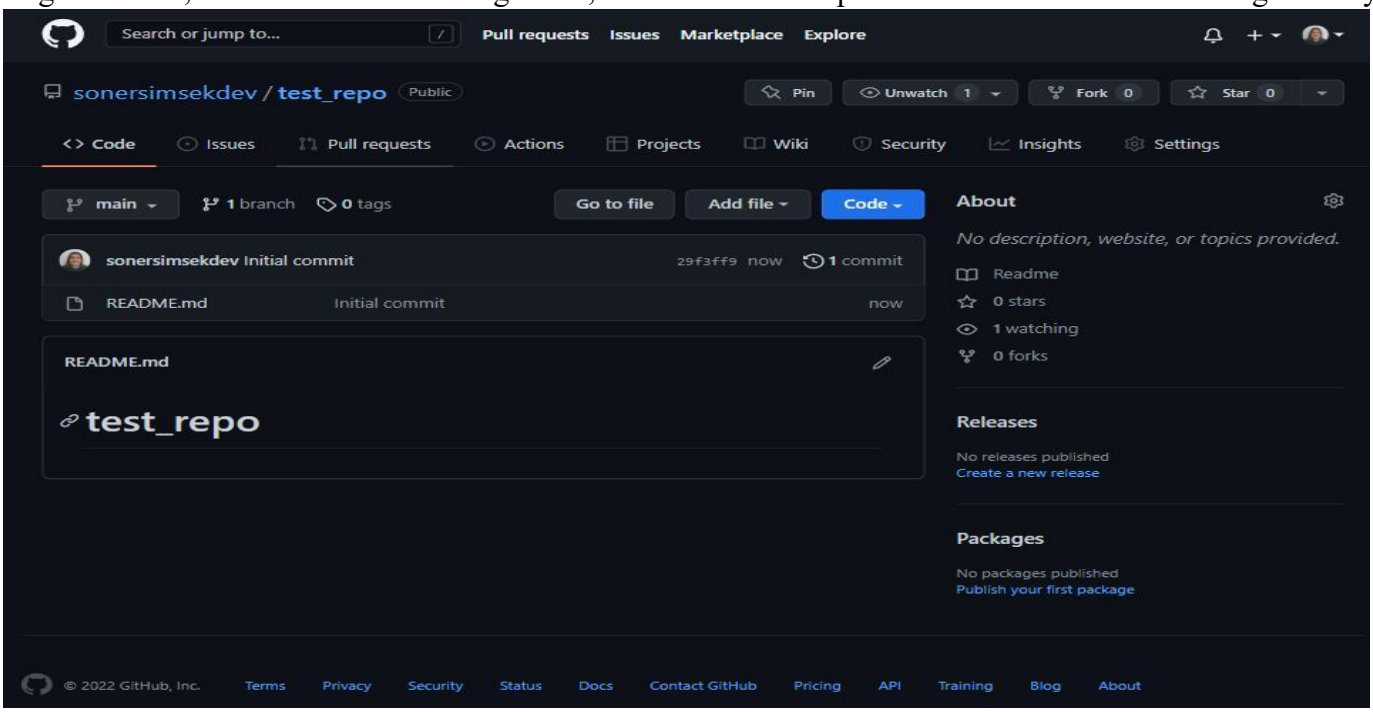
Definition
GitHub is a cloud-based platform that uses **Git** for version control. It allows multiple developers to work on the same project, store source code, and track changes.

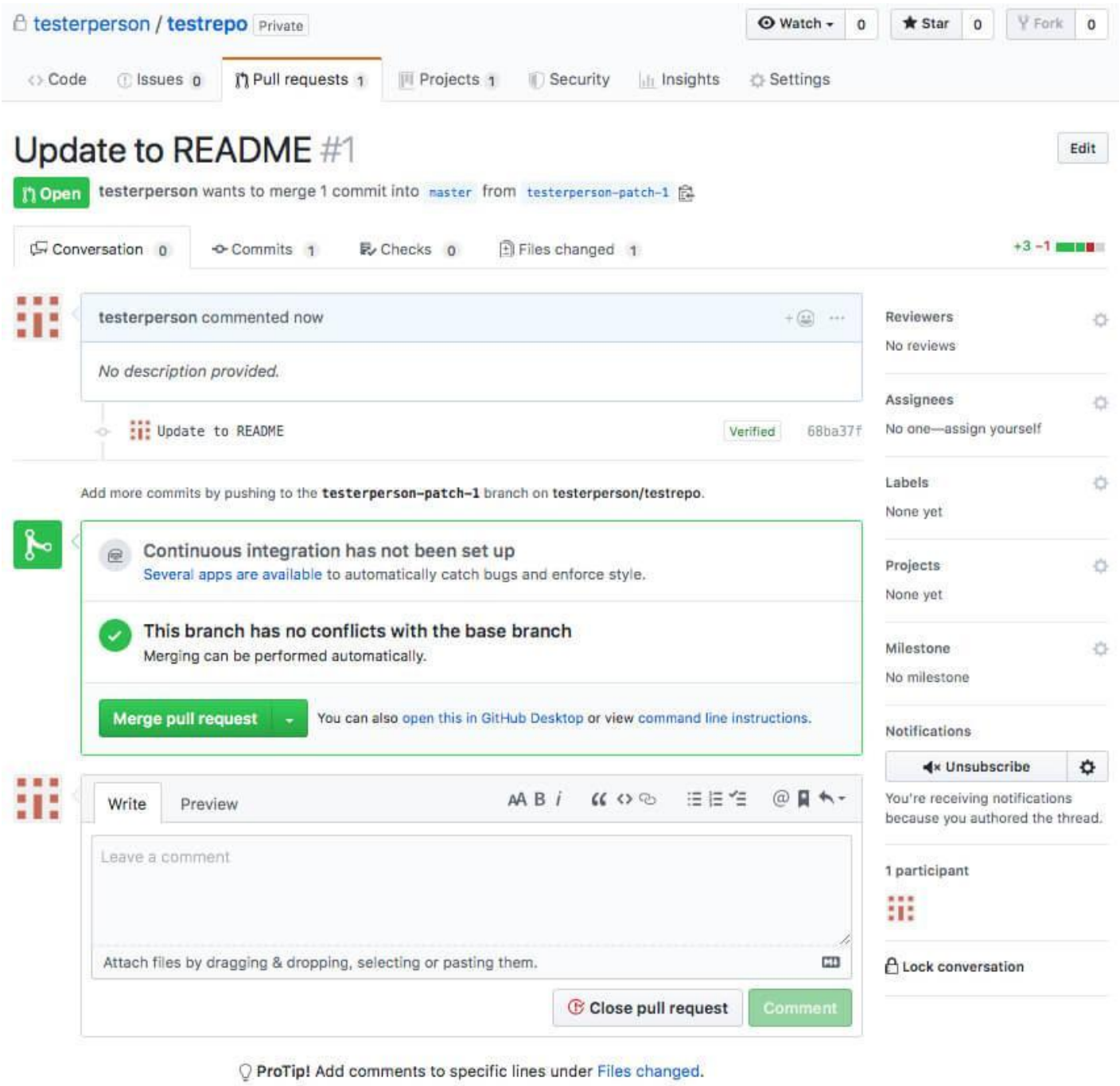
Features
- Version Control
- Branch Management
- Pull Requests
- Code Review
- Issue Tracking
- Repository Hosting

Advantages
- Prevents code loss.
- Supports teamwork.
- Maintains complete code history.
- Easy collaboration among developers.

Disadvantages
- Beginners may find Git commands difficult.
- Public repositories are visible unless using private repositories.

Real-World Example
A team building a **Library Management System** stores the project on GitHub. One developer works on the Login module, another on Book Management, and another on Reports. GitHub combines all changes safely.





4. StarUML

Definition

StarUML is a software modeling tool used to create **UML 2 diagrams** during the software design phase.

Features

- Use Case Diagram

- Class Diagram
- Sequence Diagram
- Activity Diagram
- Deployment Diagram
- ER Diagram Support

Advantages

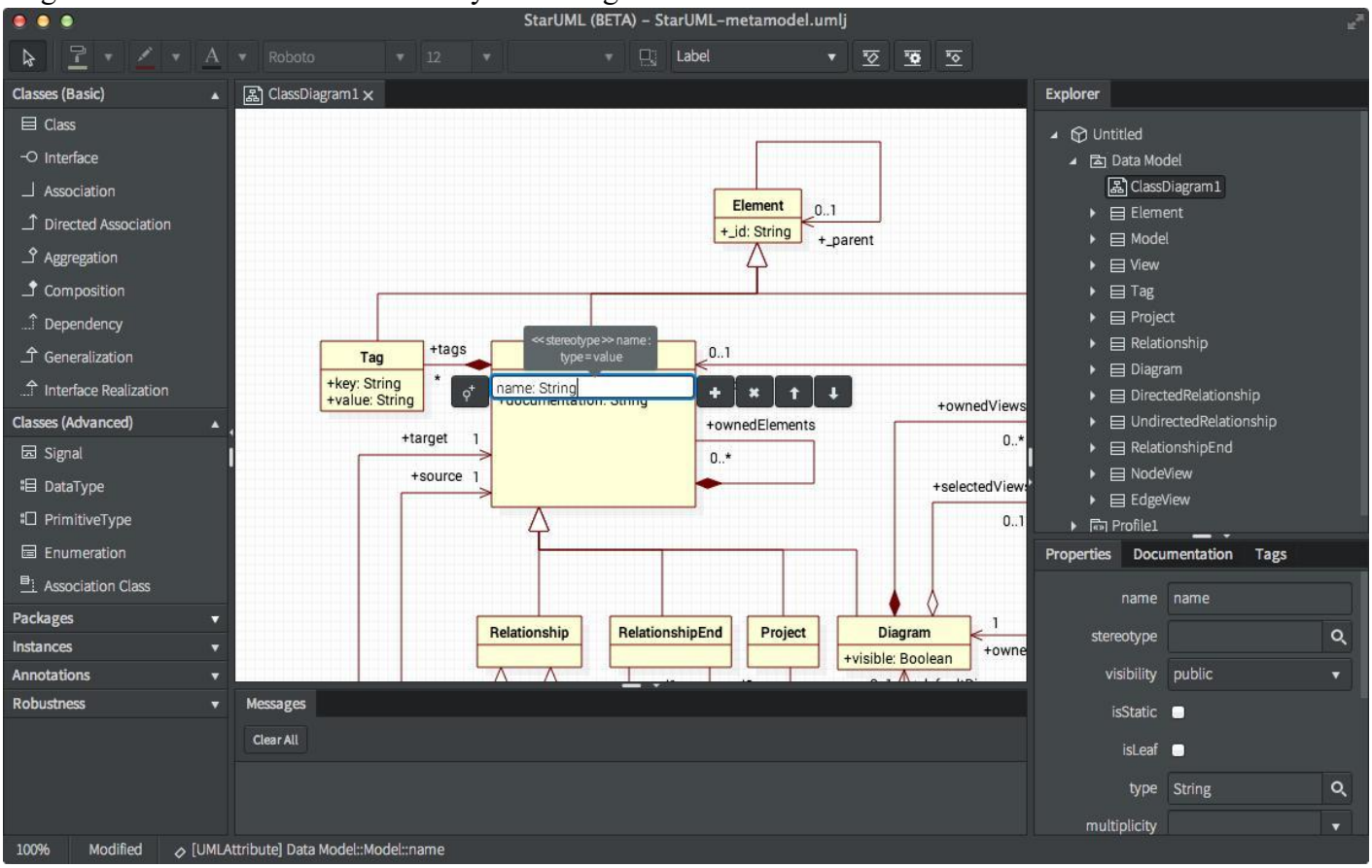
- User-friendly interface.
- Supports UML 2 standards.
- Exports diagrams to PDF and images.
- Suitable for academic and professional use.

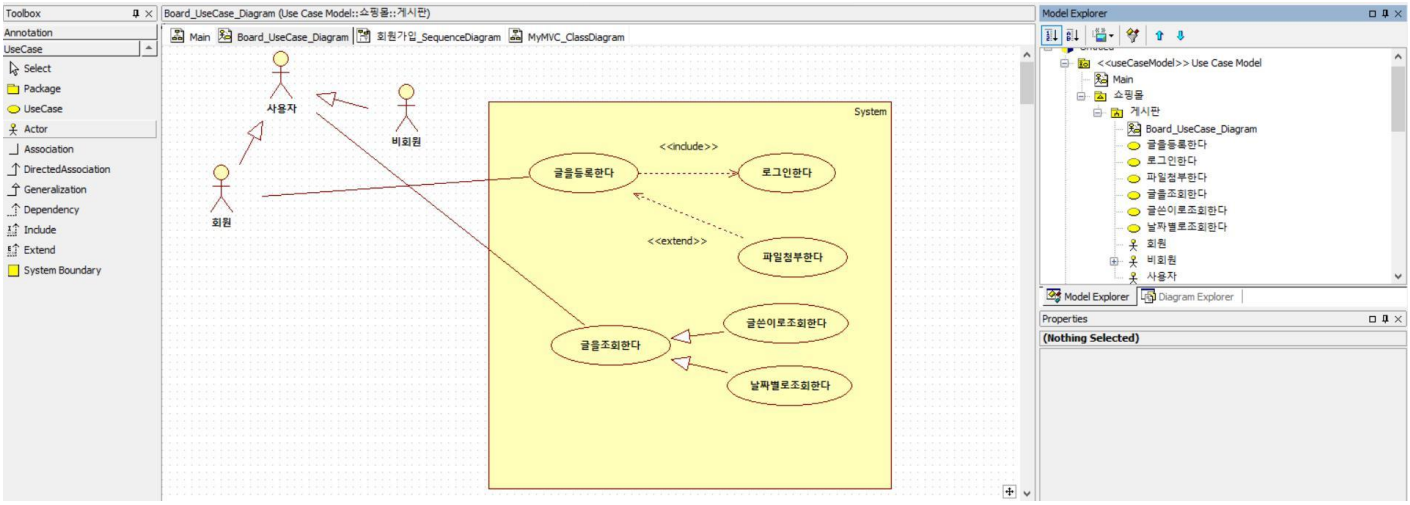
Disadvantages

- Full version is paid.
- Limited advanced features in the trial version.

Real-World Example

Before developing a **Hospital Management System**, software engineers create Use Case, Class, and Sequence diagrams in StarUML to understand system design and interactions.





5. Selenium

Definition

Selenium is an open-source automation testing tool used to test web applications automatically.

Features

- Automated Browser Testing
- Cross-Browser Testing
- Regression Testing
- Functional Testing
- Supports Java, Python, C#, and JavaScript

Advantages

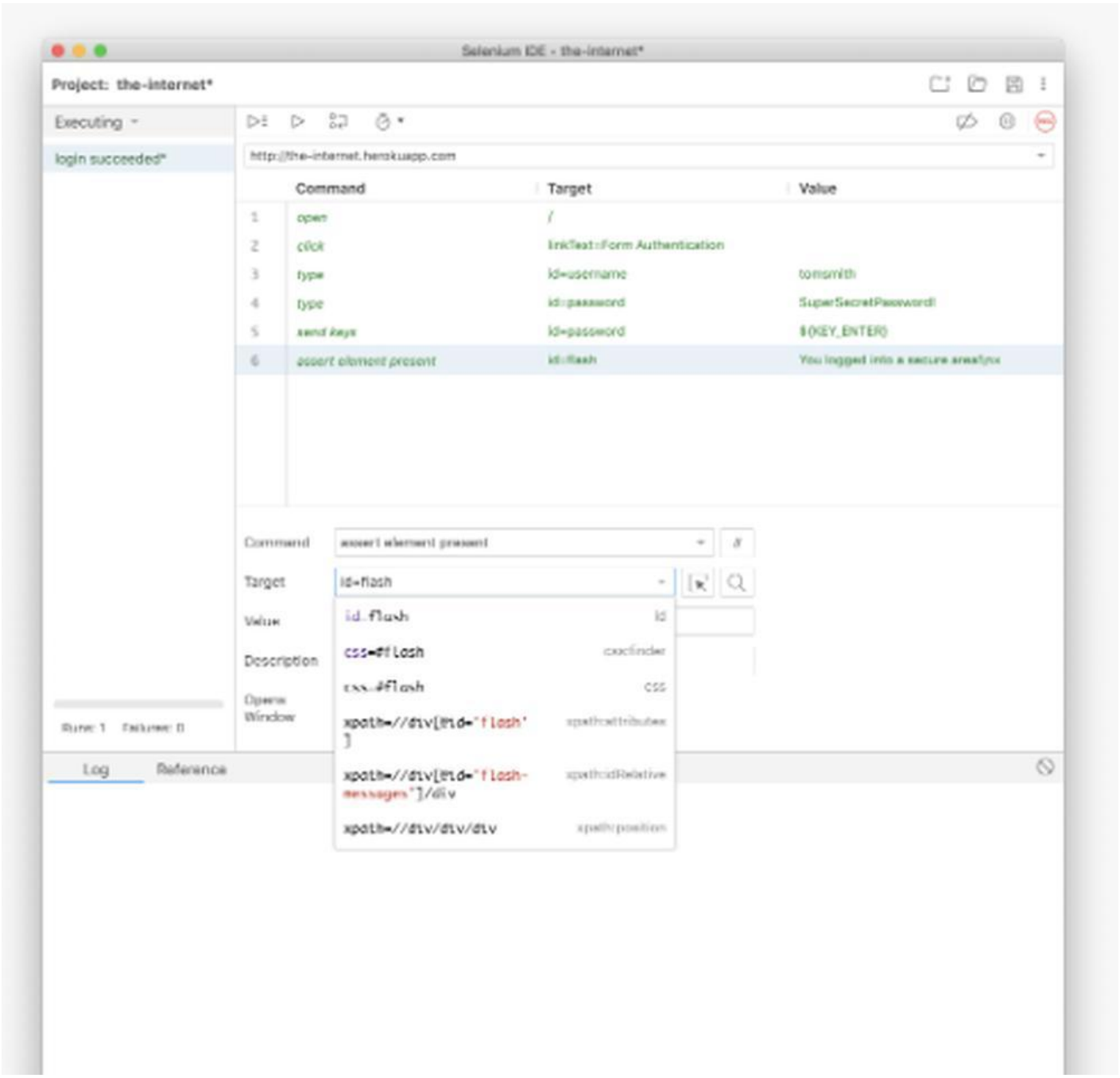
- Saves testing time.
- Reduces manual testing effort.
- Supports multiple browsers.
- Free and open source.

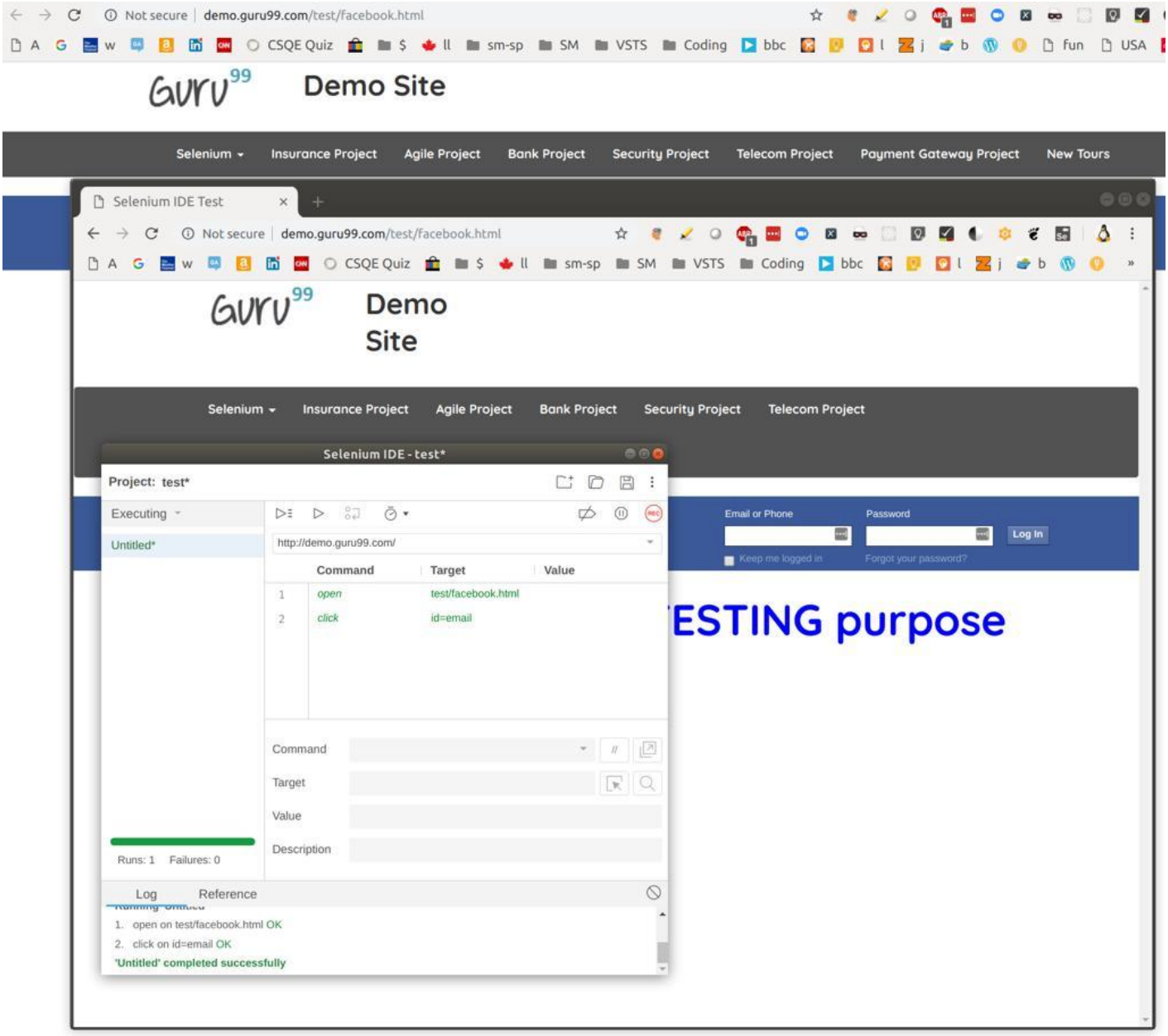
Disadvantages

- Only suitable for web applications.
- Requires programming knowledge.

Real-World Example

Before launching an **Online Banking System**, Selenium automatically tests the Login, Fund Transfer, and Payment pages across Chrome, Firefox, and Edge.


Selenium IDE - the-internet*
Project: the-internet*
Executing
login succeeded*
http://the-internet.herokuapp.com
Command
Target
Value
1 open /
2 click linkText=Form Authentication
3 type id=username tomsmith
4 type id=password SuperSecretPassword!
5 send keys id=password ${KEY_ENTER}
6 assert element present id=flash You logged into a secure area!
Command assert element present
Target id=flash
Value
Description
Opens Window
id=flash id
css=#flash cssFinder
css=#flash css
xpath=//div[@id='flash'] xpathFullAttributes
xpath=//div[@id='flash-messages']/div xpathRelative
xpath=//div/div/div xpathPosition
Runs: 1 Failures: 0
Log
Reference



6. Docker

Definition

Docker is a containerization platform that packages an application along with all its dependencies into a **container**, ensuring it runs consistently on different systems.

Features

- Application Containerization
- Fast Deployment
- Environment Consistency

- Lightweight Containers
- Easy Scaling

Advantages

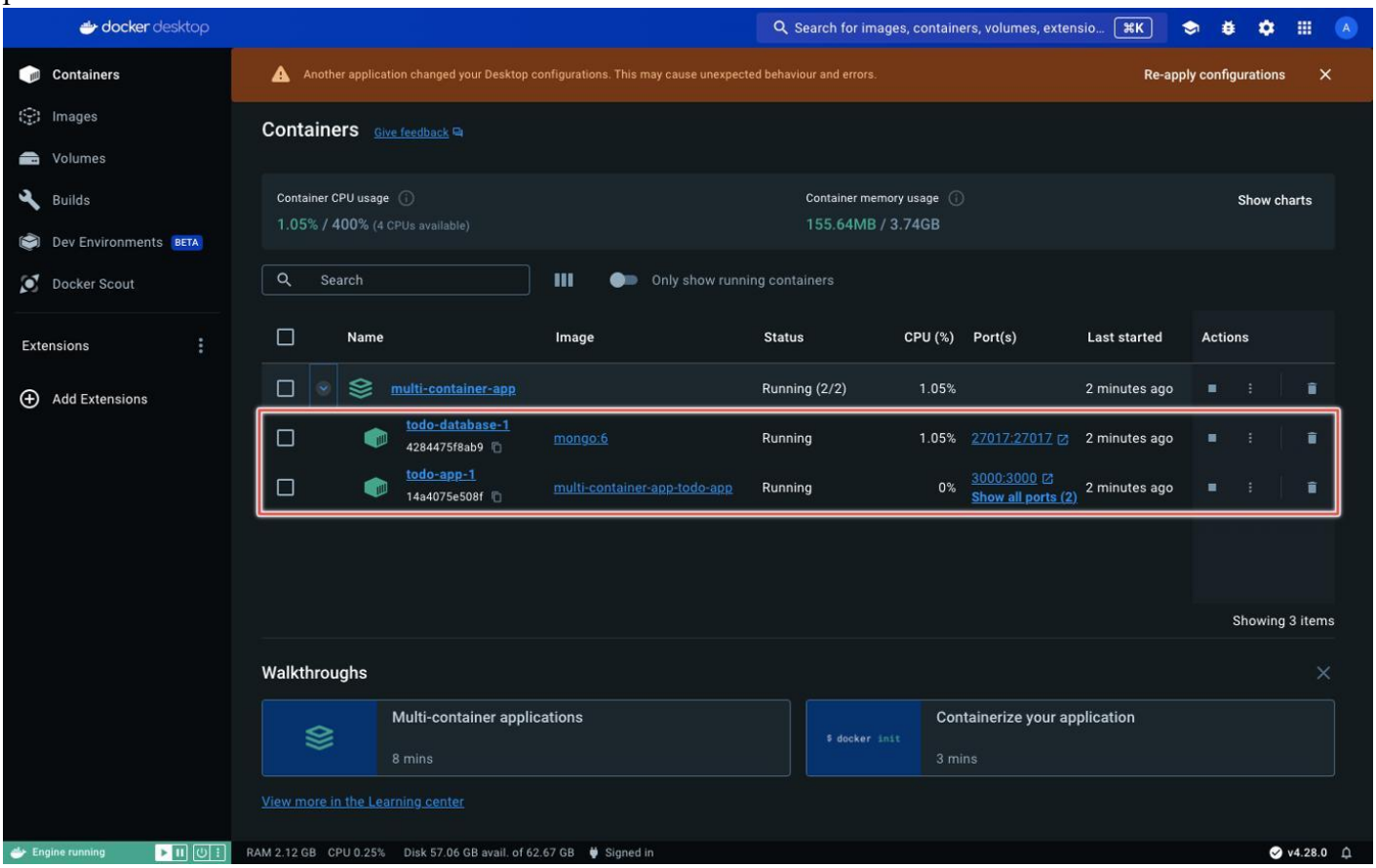
- Eliminates "works on my machine" issues.
- Simplifies deployment.
- Improves portability.
- Uses fewer resources than virtual machines.

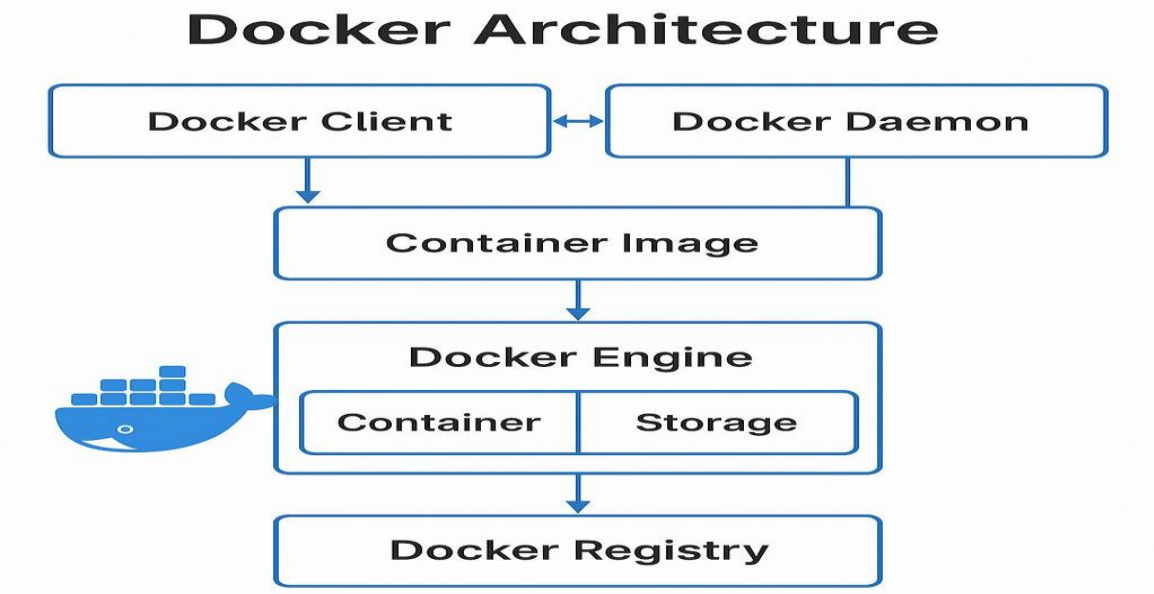
Disadvantages

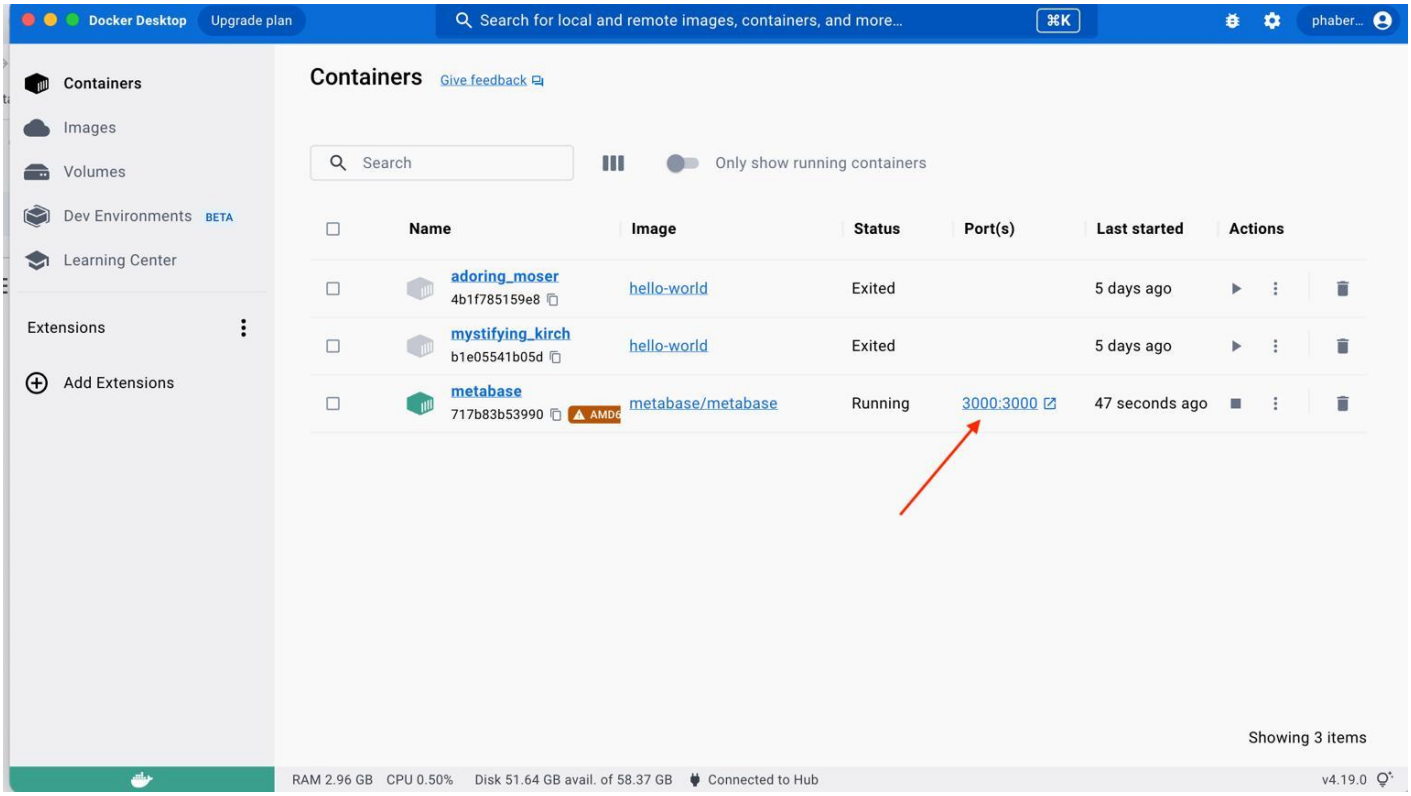
- Initial setup can be challenging.
- Requires knowledge of containers and images.

Real-World Example

A company developing a **Music Streaming Application** package the backend, frontend, and database using Docker containers. The application runs the same way on developers' computers, testing servers, and production servers.







➢ Testing Tools

1. Junit

Definition

JUnit is a unit testing framework for Java used to test individual methods and classes.

Features

- Unit testing

- Assertions
- Test annotations
- Test reports
- Automatic execution

Advantages

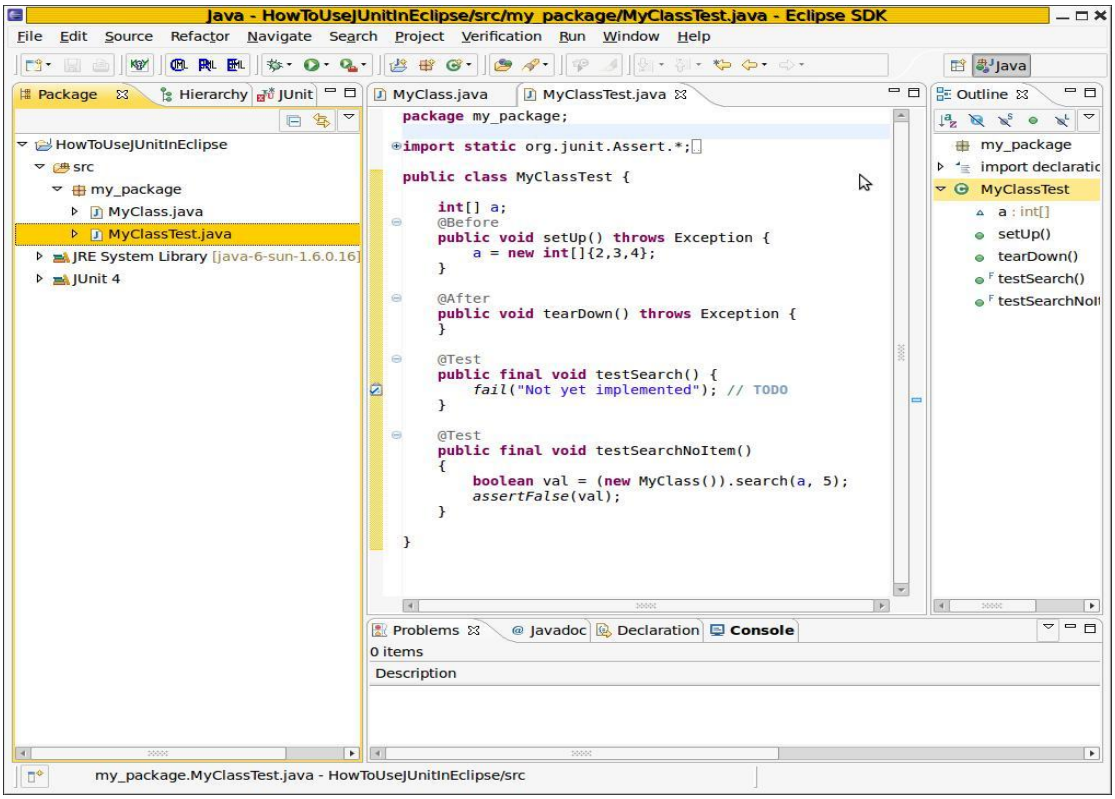
- Easy to write tests
- Detects bugs early
- Integrates with Maven and Jenkins

Disadvantages

- Only for Java
- Does not test user interfaces

Real-World Example

A banking application uses JUnit to verify that the calculateInterest() function returns the correct value.



2. TestNG

Definition

TestNG is an advanced testing framework for Java that supports unit, functional, and integration testing.

Features

- Parallel testing
- Test grouping
- Data-driven testing
- HTML reports

- Flexible annotations

Advantages

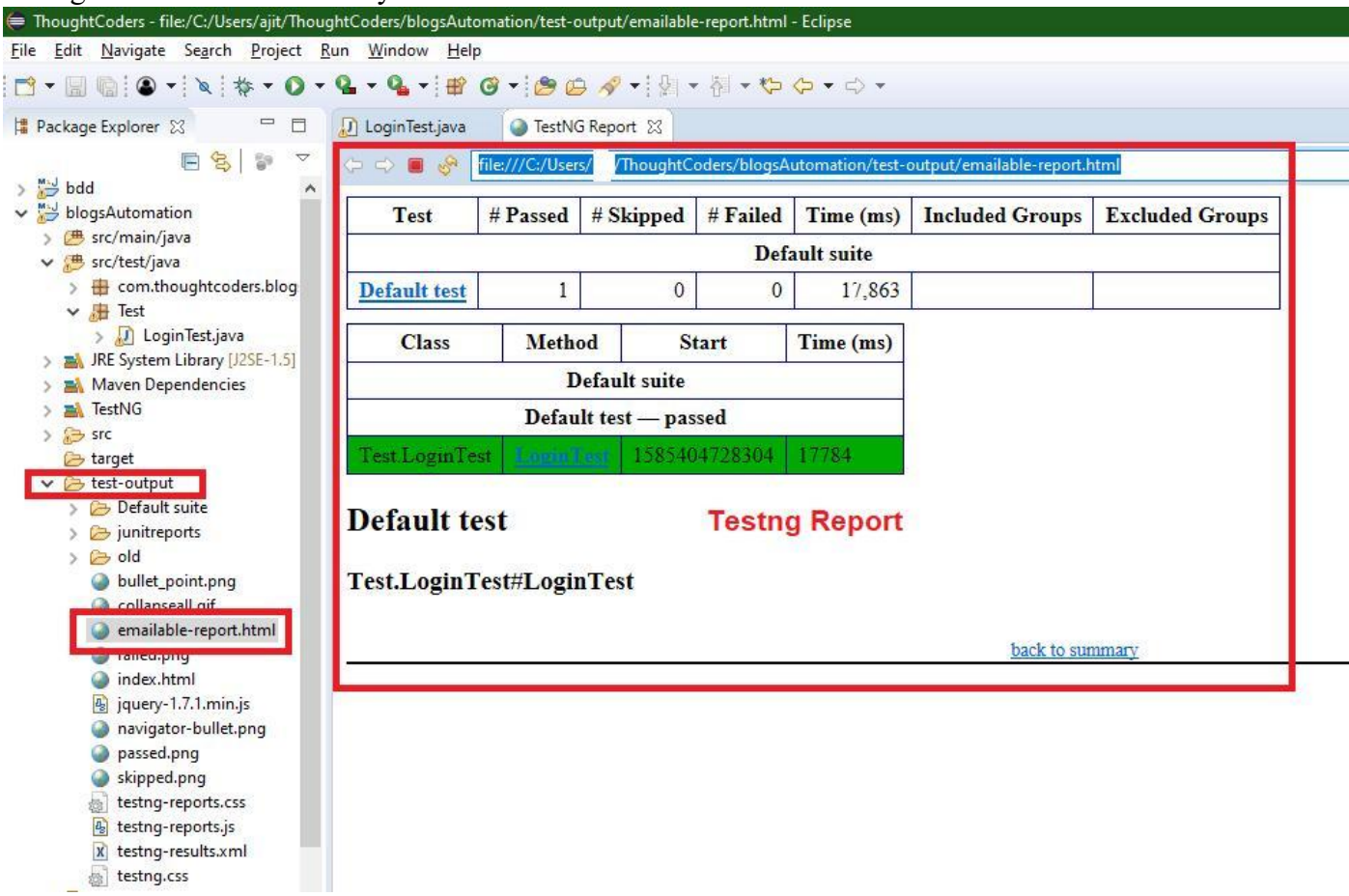
- Faster execution
- Better reporting
- Supports multiple test cases together

Disadvantages

- Mainly used with Java
- More complex than JUnit

Real-World Example

A **Hospital Management System** uses TestNG to test patient registration, appointment booking, and billing modules simultaneously.



3.Postman

Definition

Postman is an API testing tool used to test REST and SOAP web services.

Features

- API request testing
- Authentication testing
- Response validation
- Automated API testing

- API documentation

Advantages

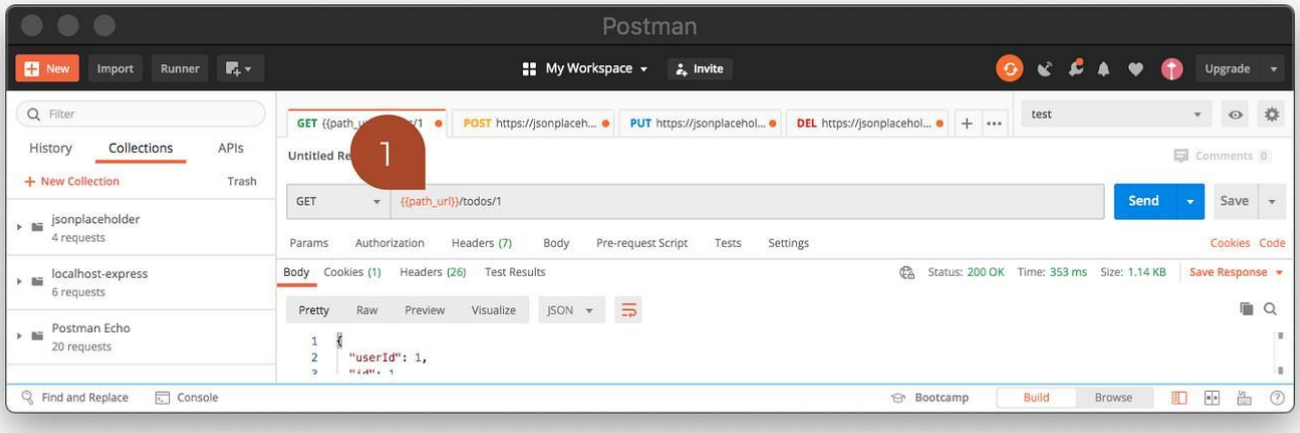
- Easy to use
- Supports REST APIs
- Fast API debugging

Disadvantages

- Not suitable for UI testing
- Limited performance testing

Real-World Example

A Music Streaming App uses Postman to test APIs such as Login API, Search Song API, and Playlist API.



4.Apache JMeter

Definition

Apache JMeter is an open-source tool used for **performance, load, and stress testing**.

Features

- Load testing
- Stress testing
- Performance analysis
- Graphical reports
- Supports web applications and APIs

Advantages

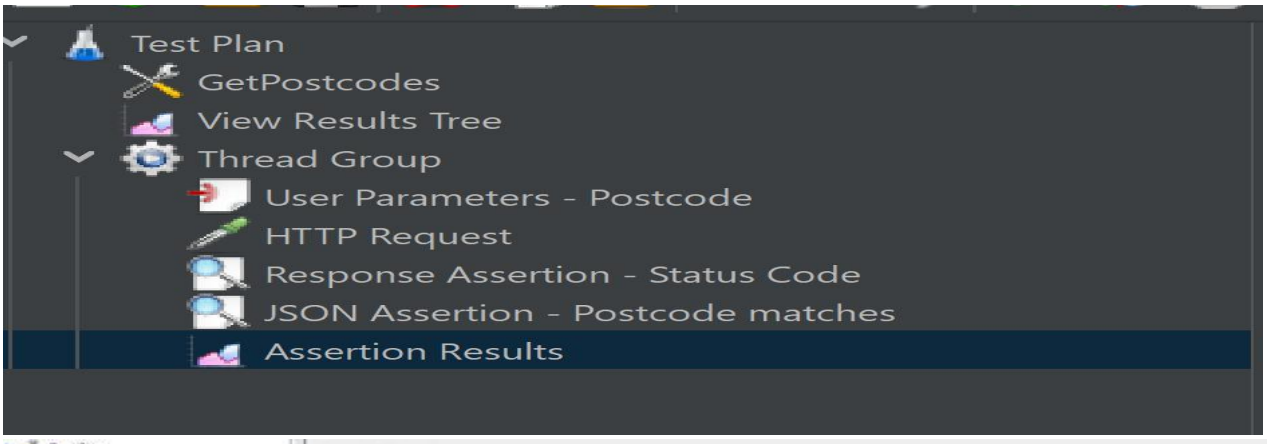
- Free
- Supports multiple protocols
- Generates detailed performance reports

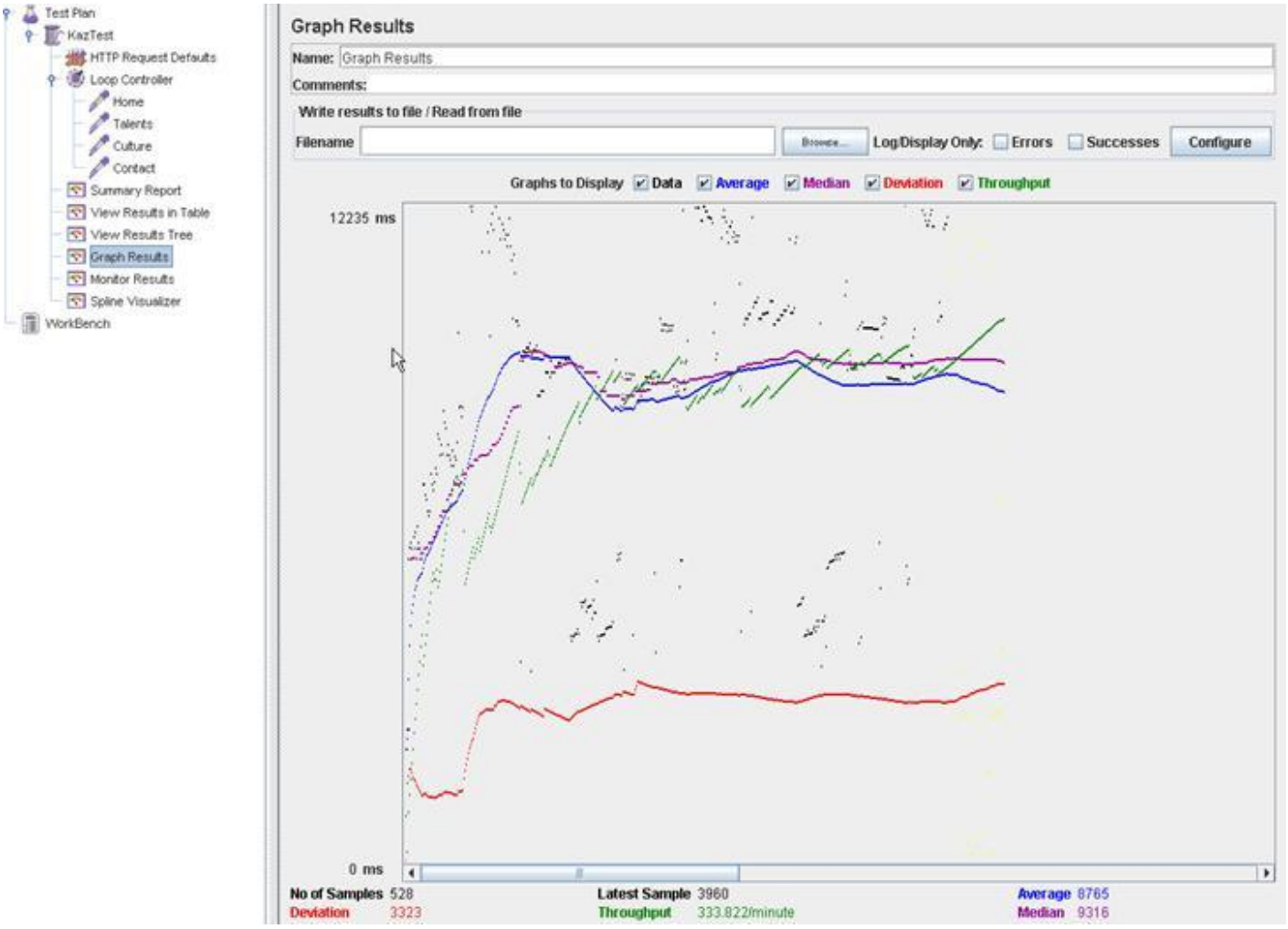
Disadvantages

- Requires configuration
- High load tests consume more system resources

Real-World Example

An **Online Shopping System** uses JMeter to check whether the website can handle **10,000 users** during a sale.





5. Appium

Definition

Appium is an open-source tool used for **mobile application testing** on Android and iOS devices.

Features

- Mobile automation
- Cross-platform testing
- Supports native and hybrid apps
- Works with multiple programming languages

Advantages

- Free
- Tests Android and iOS apps
- Reuses automation scripts

Disadvantages

- Slower than native testing tools
- Requires mobile device setup

Real-World Example

A **Food Delivery App** uses Appium to automate login, food ordering, and payment testing on Android and iPhone devices

